

CSE 4621: Machine Learning

Linear Regression

Lecture 3

Ishmam Tashdeed
Lecturer, CSE IUT

Lecture Outline

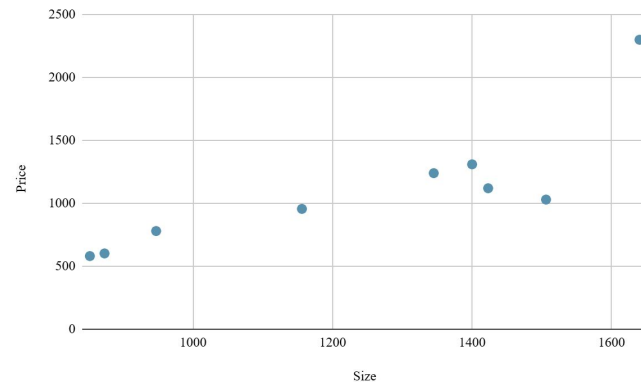
- Formulating the Problem
 - Motivations
 - Problem Formulation
 - Hypothesis Function
 - Notation
- Cost Function
 - Mean Squared Error
 - Contour Plots
 - Geometric Interpretation
 - Convexity of MSE
- OLS Derivation
 - Normal Equations
 - Matrix Form
 - Geometric view
 - Closed-form Solution
- Gradient Descent
 - Learning Rate
 - Convergence Analysis
 - Batch/Stochastic/Mini-batch GD
- Extensions

What is Linear Regression?

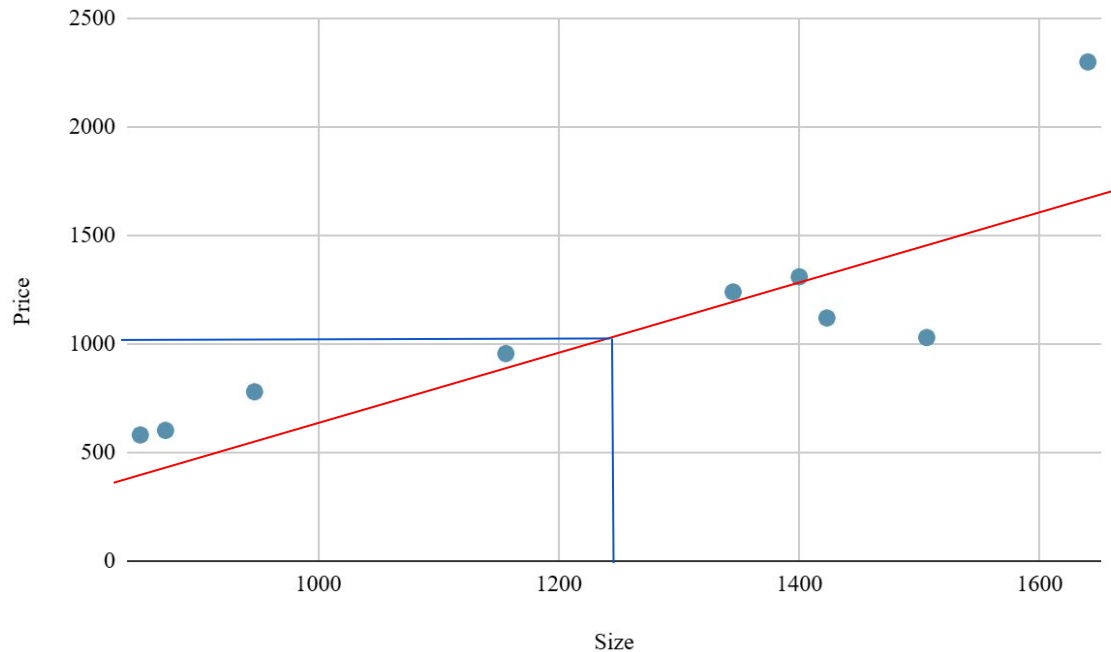
Regression is a *supervised learning* task where we predict a continuous / real-valued output

Suppose you want to buy a house in Dhaka and the only thing that mattered for the price was the size of the house.

Size (sq. feet)	Price (\$)
1506	1030
1640	2300
...	...
852	581



What is Linear Regression?



What is Linear Regression?

Given the input features x , the goal is to learn $h : x \rightarrow y$ where f is a **weighted sum** of different values (as h is supposed to be a linear function)

Notation:

m = Number of training examples

x = Input variables / features

y = Output variable

$(x^{(i)}, y^{(i)}) = i^{\text{th}}$ training example

Size (sq. feet) (x)	Price (\$) (y)
1506	1030
1640	2300
...	...
852	581

What is Linear Regression?

Training Data: $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^m$

Hypothesis function: $h_{\theta}(x) = \theta_0 + \theta_1 x$

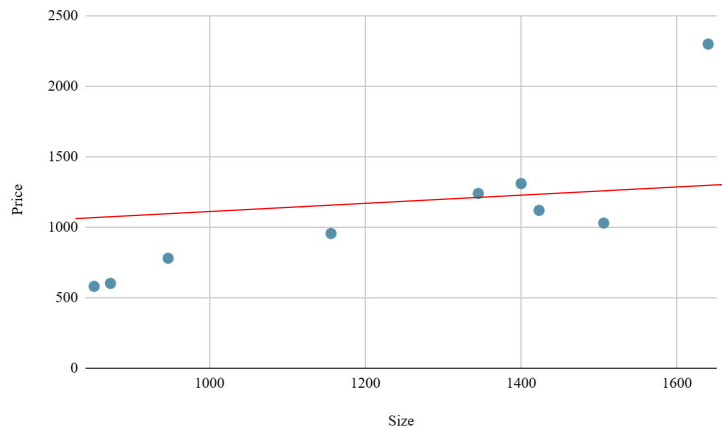
θ_i = Parameters

How can we represent this through linear algebra?

How can we choose the right θ_i values?

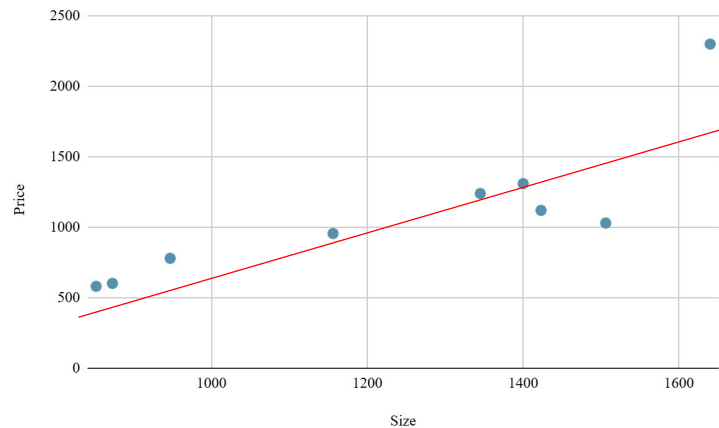
Size (sq. feet) (x)	Price (\$) (y)
1506	1030
1640	2300
...	...
852	581

What is Linear Regression?



$$\theta_0 = 1010$$

$$\theta_1 = 1.1$$



$$\theta_0 = 483$$

$$\theta_1 = 2.3$$

Which one is better?

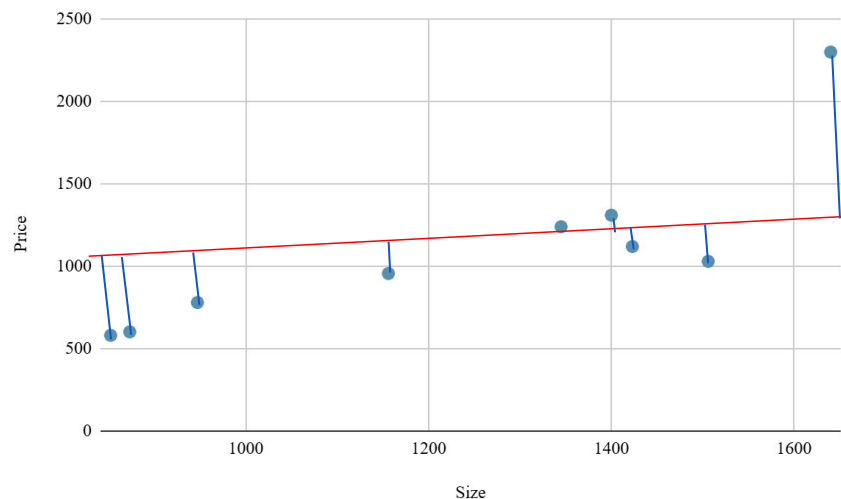
Cost Function

Residual for the i^{th} example: $e^{(i)} = y^{(i)} - \hat{y}^{(i)} = y^{(i)} - (\theta_0 + \theta_1 x)$

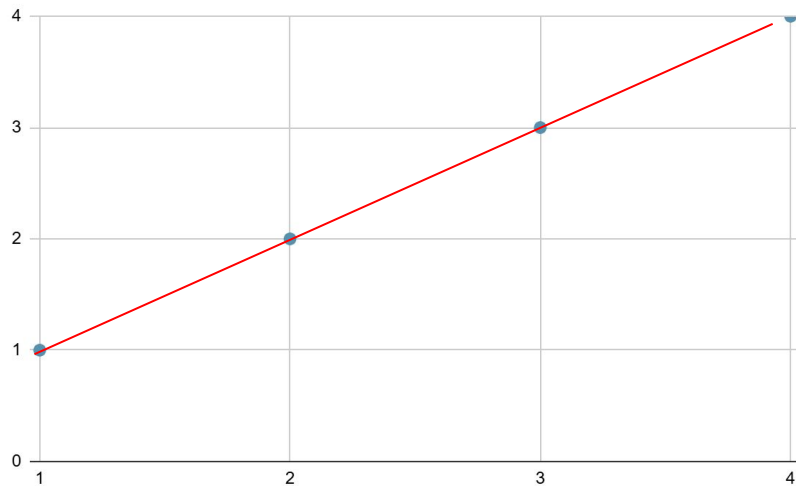
Mean Squared Error (MSE) loss: $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (y^{(i)} - (\theta_0 + \theta_1 x))^2$

The $(1/2)$ is a convention that simplifies the gradient descent process

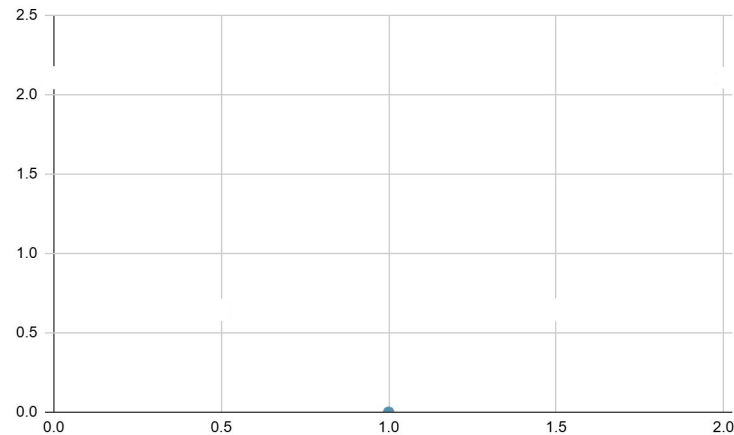
The goal for any cost function is to minimize the loss/cost for all θ



Cost Function

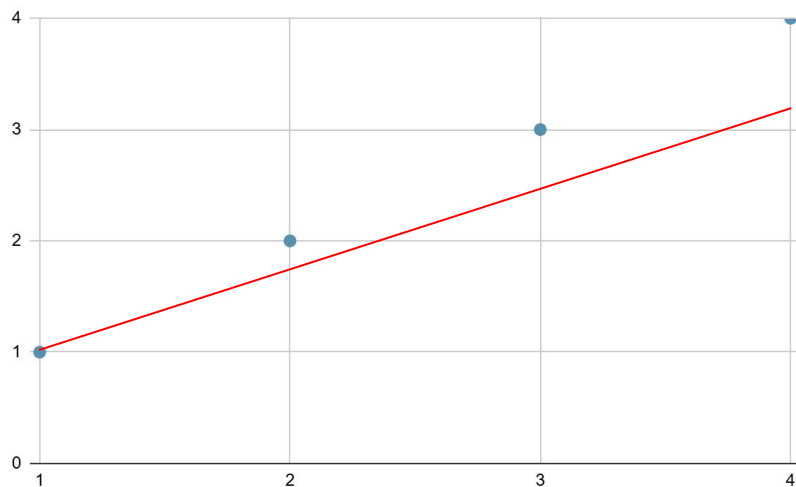


$h_{\theta}(x)$

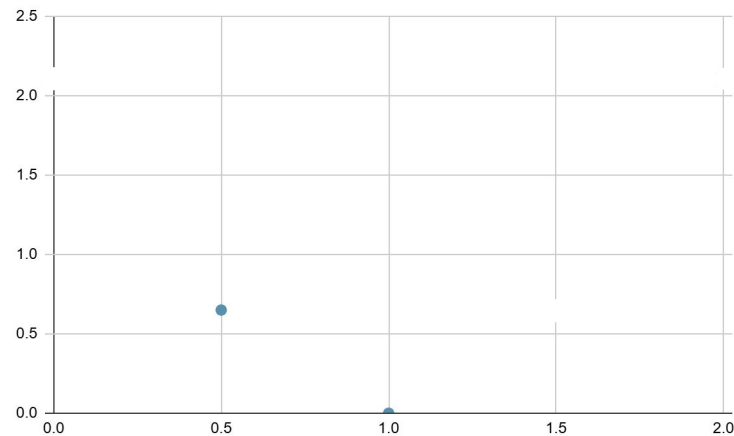


$J(\theta)$

Cost Function

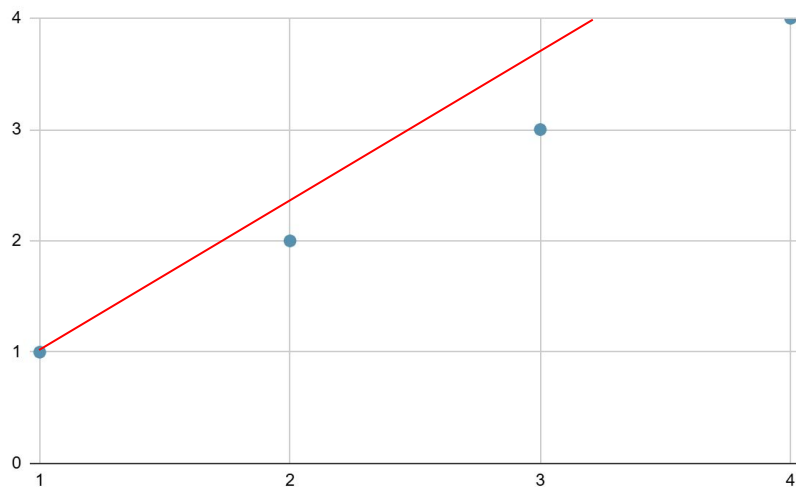


$h_{\theta}(x)$

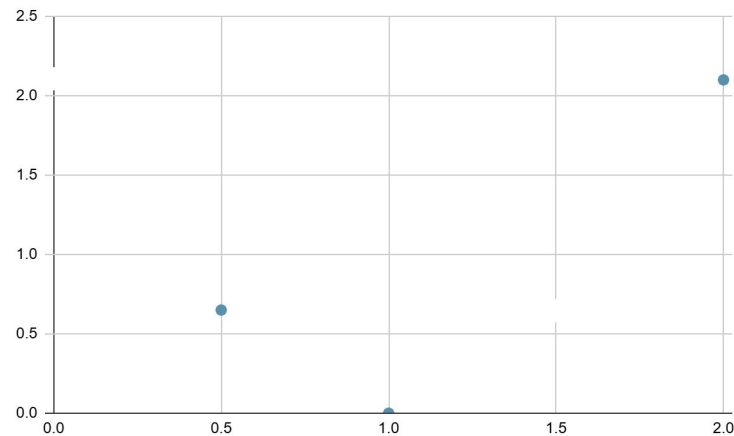


$J(\theta)$

Cost Function

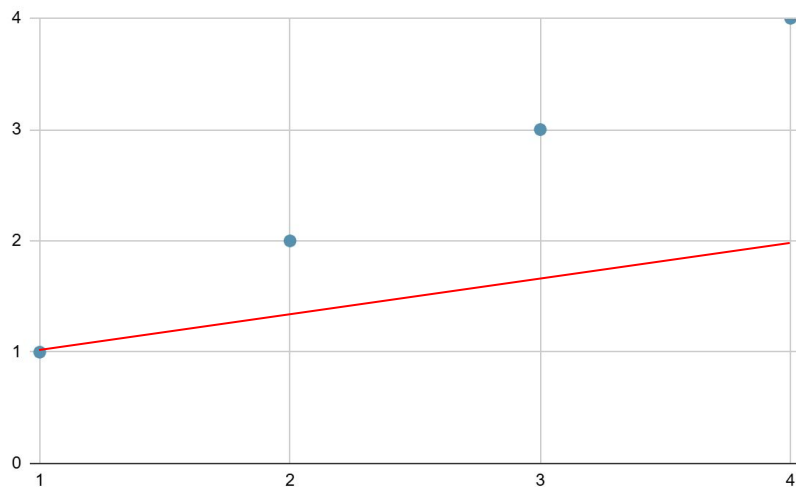


$h_{\theta}(x)$

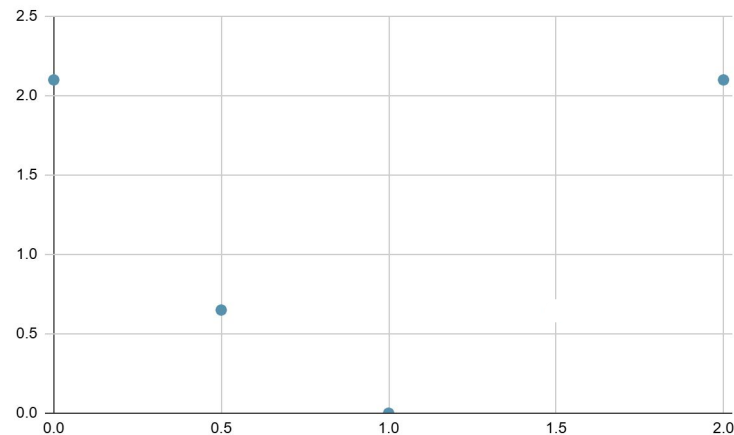


$J(\theta)$

Cost Function

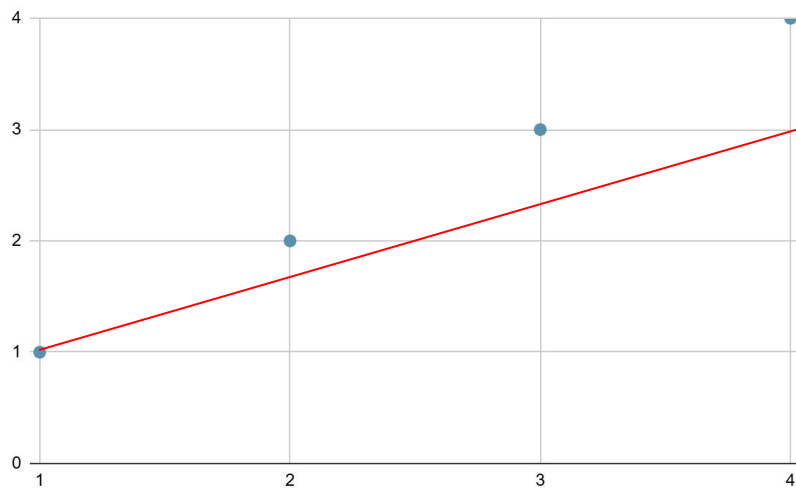


$h_{\theta}(x)$

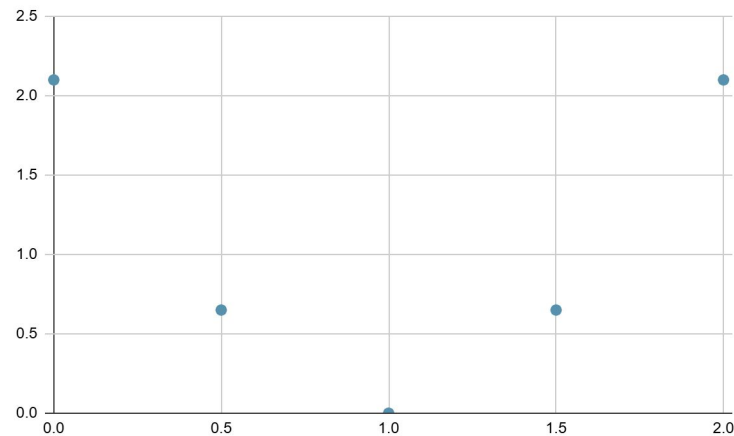


$J(\theta)$

Cost Function



$h_{\theta}(x)$

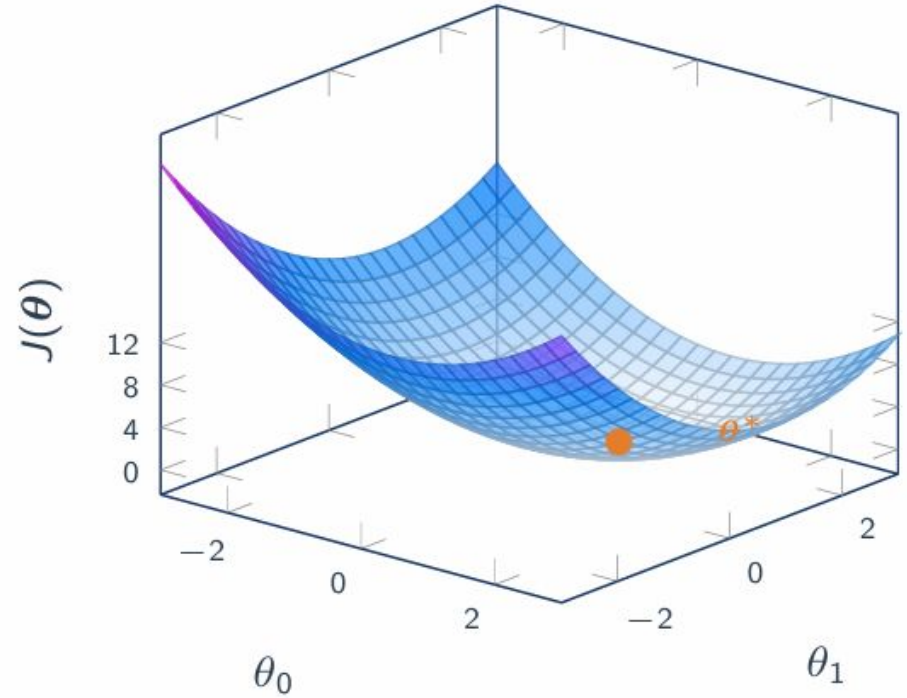
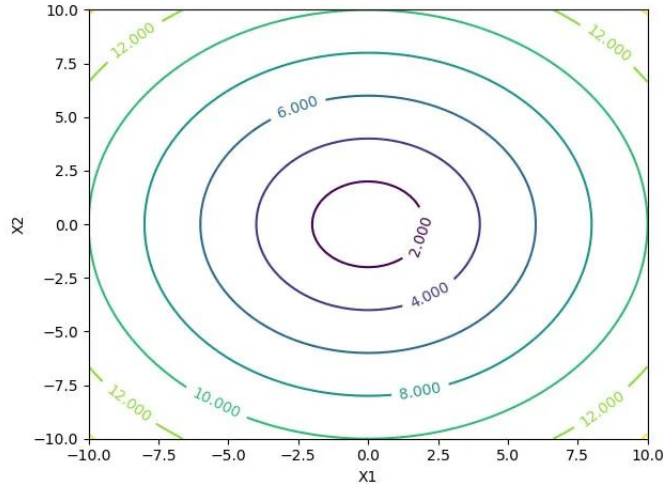


$J(\theta)$

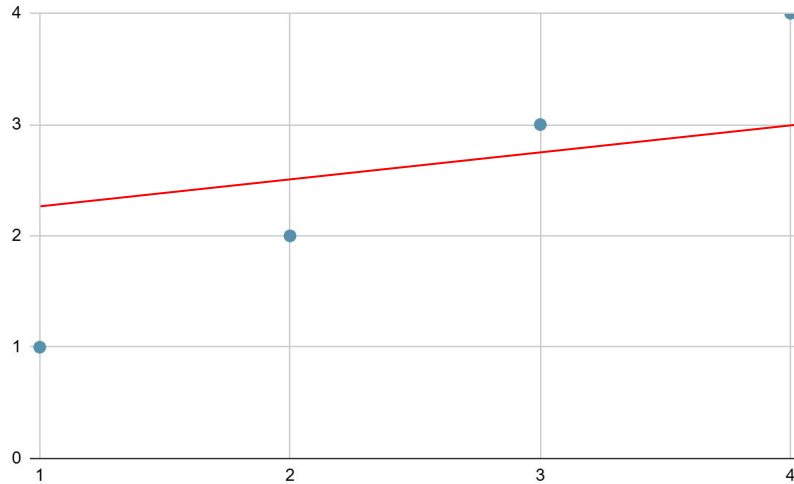
Cost Function

For $J(\theta_0, \theta_1)$, it will be a convex structure:

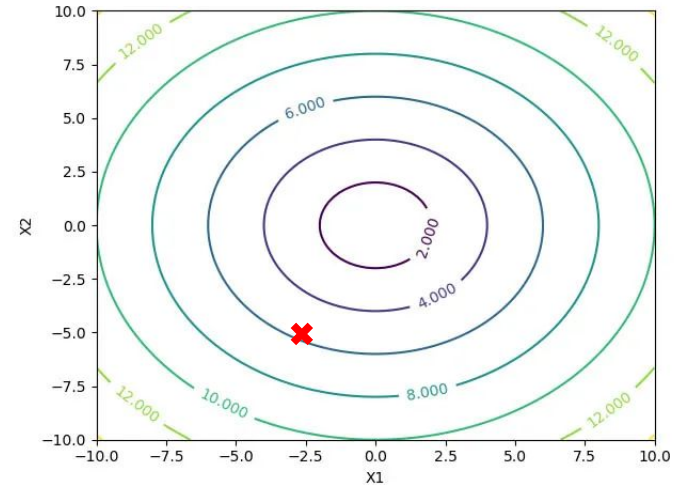
The same structure can be represented by a contour plot



Cost Function

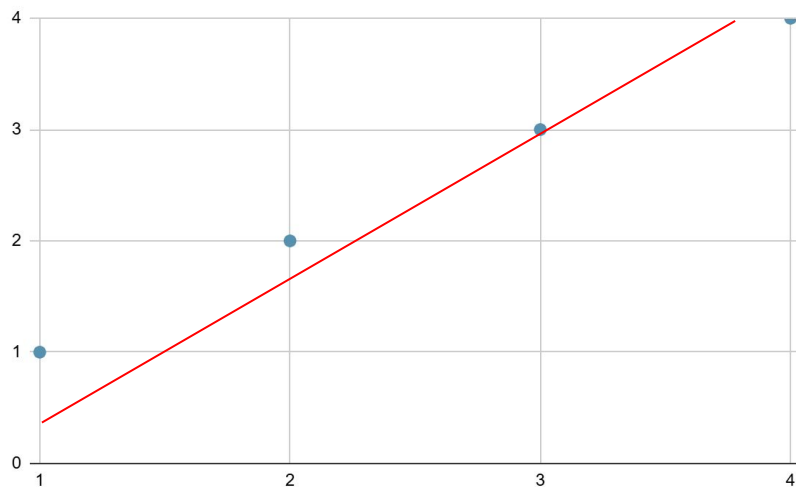


$$h_{\theta}(x)$$

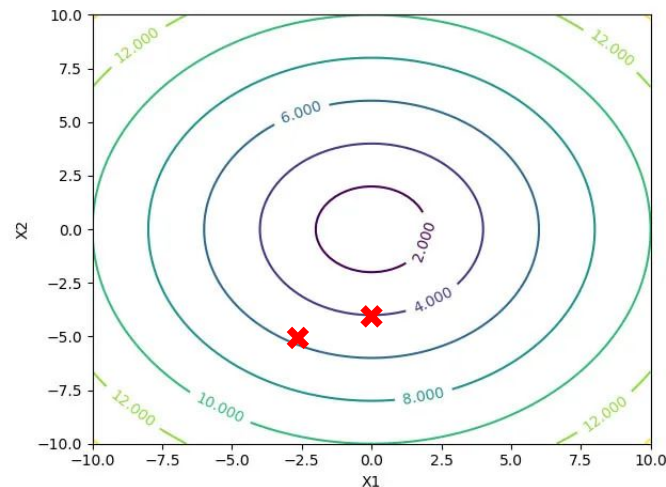


$$J(\theta)$$

Cost Function

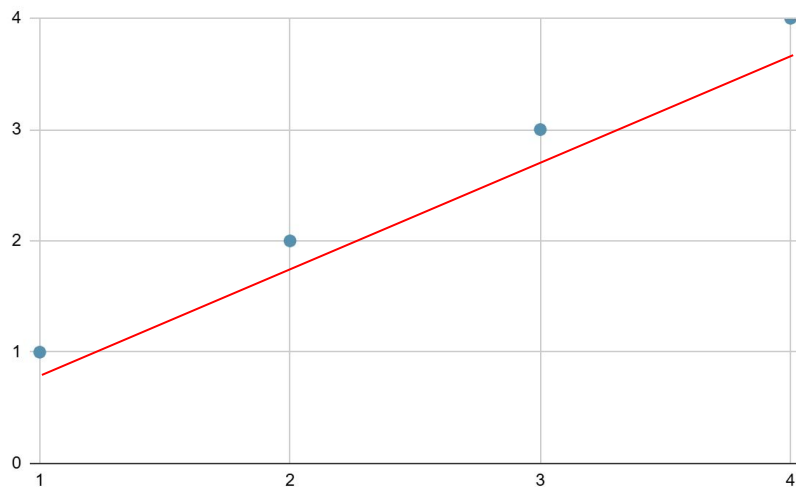


$$h_{\theta}(x)$$

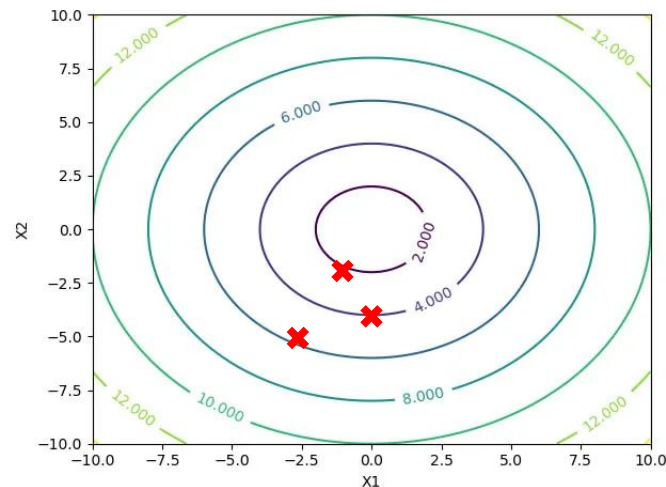


$$J(\theta)$$

Cost Function

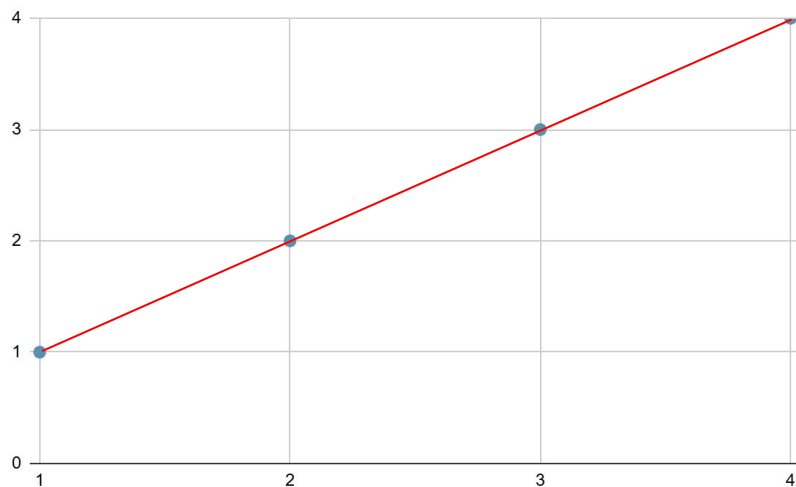


$$h_{\theta}(x)$$

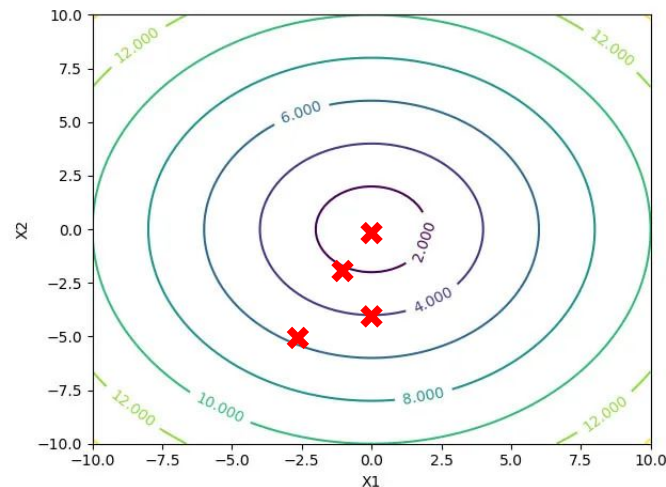


$$J(\theta)$$

Cost Function

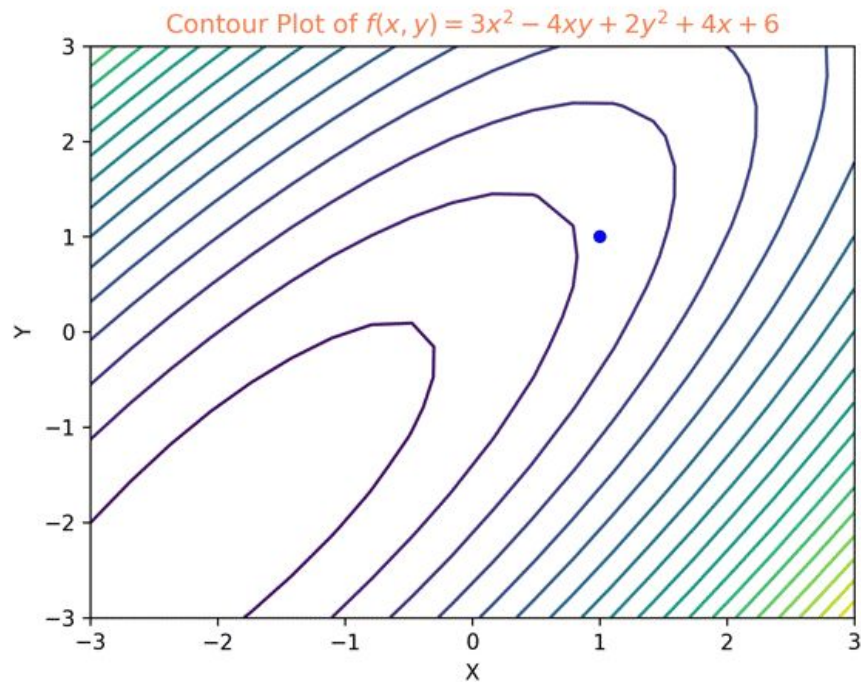


$$h_{\theta}(x)$$



$$J(\theta)$$

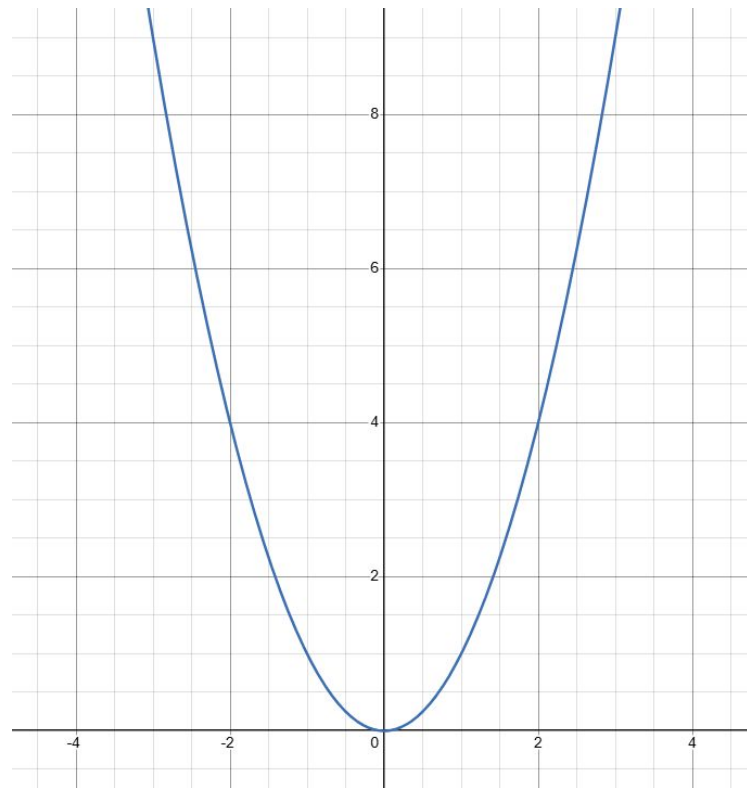
Cost Function



Gradient Descent

- Start with some random θ_0, θ_1
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we reach a minimum value

But how to change θ_0, θ_1 ?

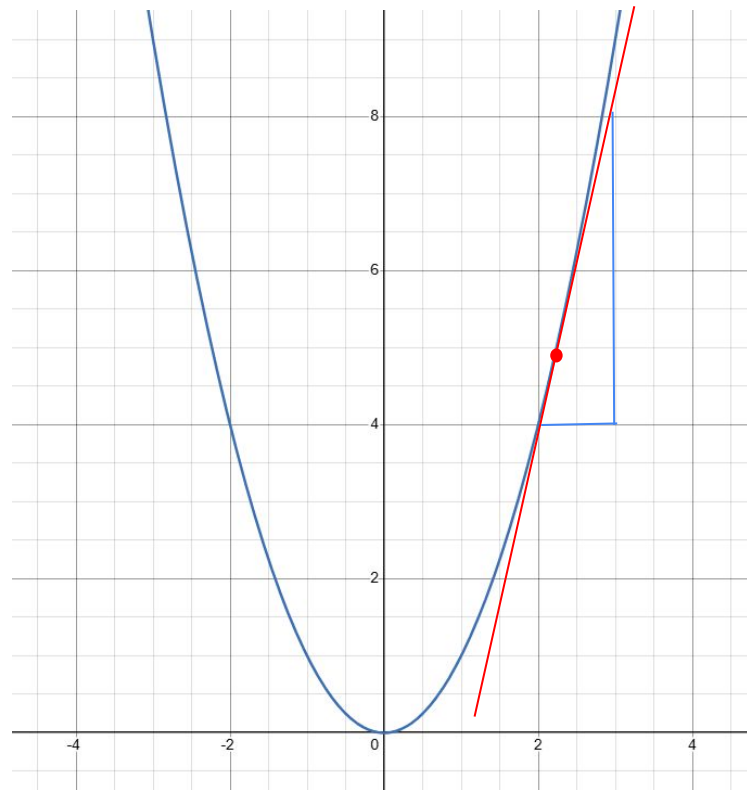


Gradient Descent

- Start with some random θ_0, θ_1
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we reach a minimum value

But how to change θ_0, θ_1 ?

- Take partial derivatives with respect to both the variables (θ_0, θ_1)

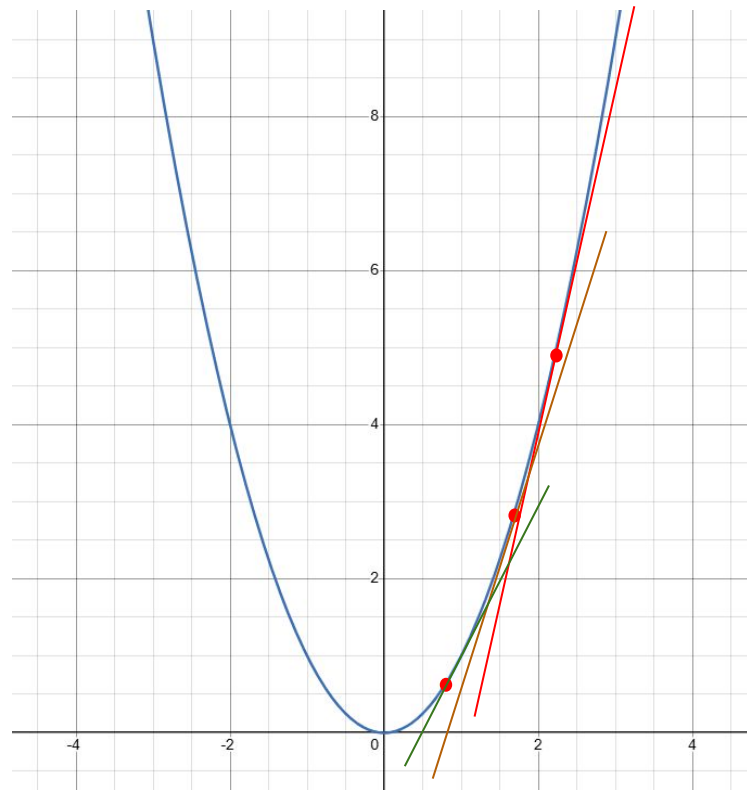


Gradient Descent

- Start with some random θ_0, θ_1
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we reach a minimum value

But how to change θ_0, θ_1 ?

- Take partial derivatives with respect to both the variables (θ_0, θ_1)
- Subtract the partial derivative from the original θ_i after scaling by a factor α (learning rate)



Gradient Descent

Thus,

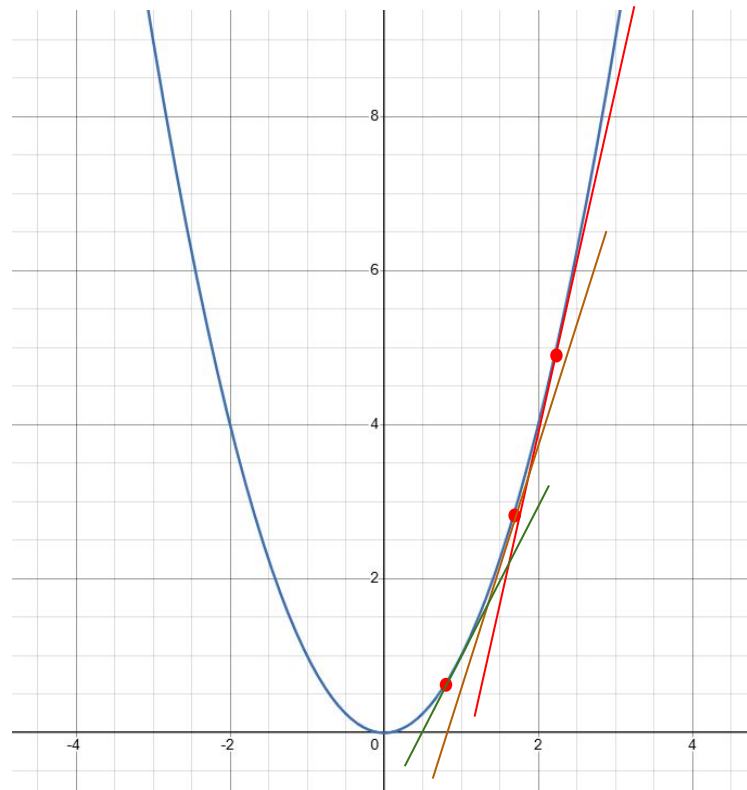
$$\theta_0 = \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_1 = \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

Where,

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$$



Gradient Descent

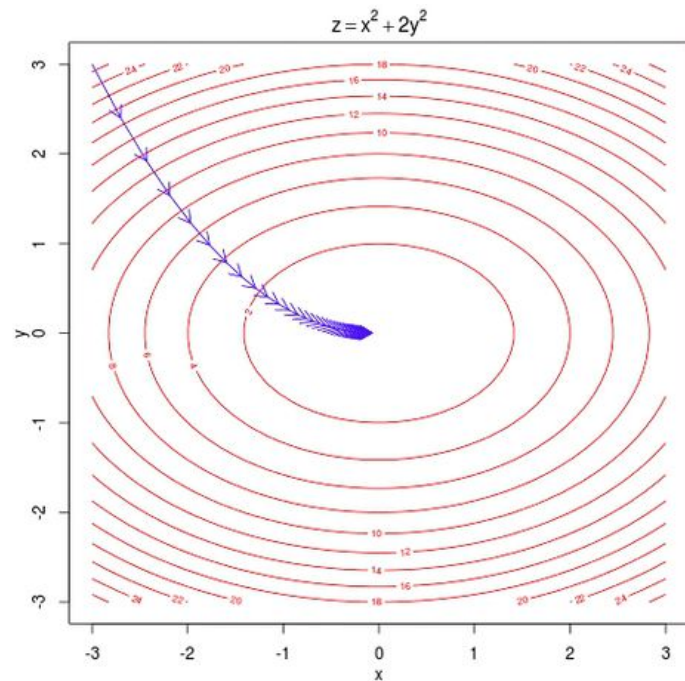
repeat until convergence {

$$\theta_0 = \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

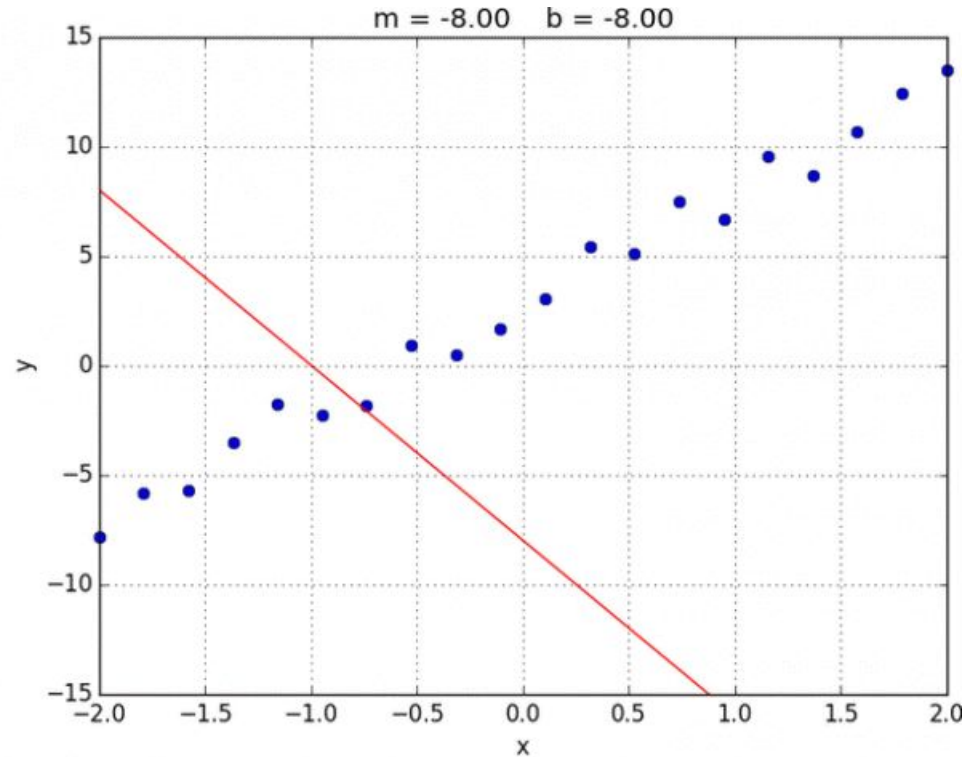
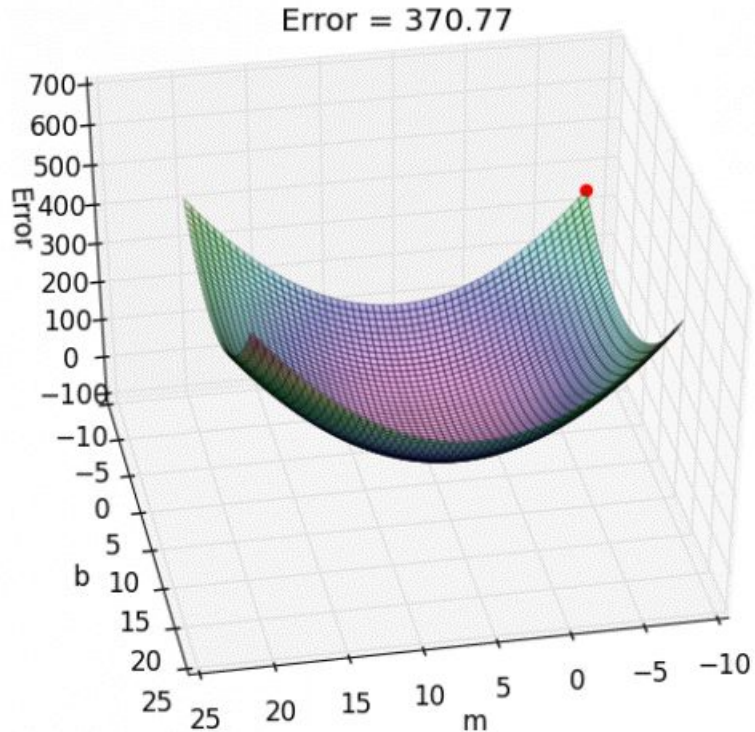
$$\theta_1 = \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

}

Update θ_0 and θ_1 simultaneously



Gradient Descent



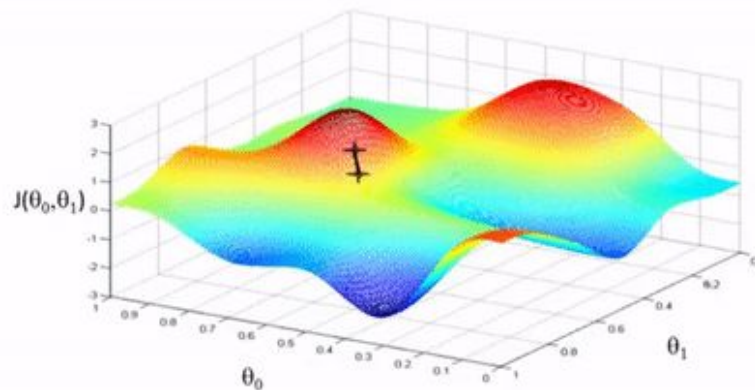
Gradient Descent

- If the learning rate (α) is too small, gradient descent can be slow
- If α is too large, gradient descent can overshoot the minimum and may fail to converge or even diverge

What if the function isn't continuous?

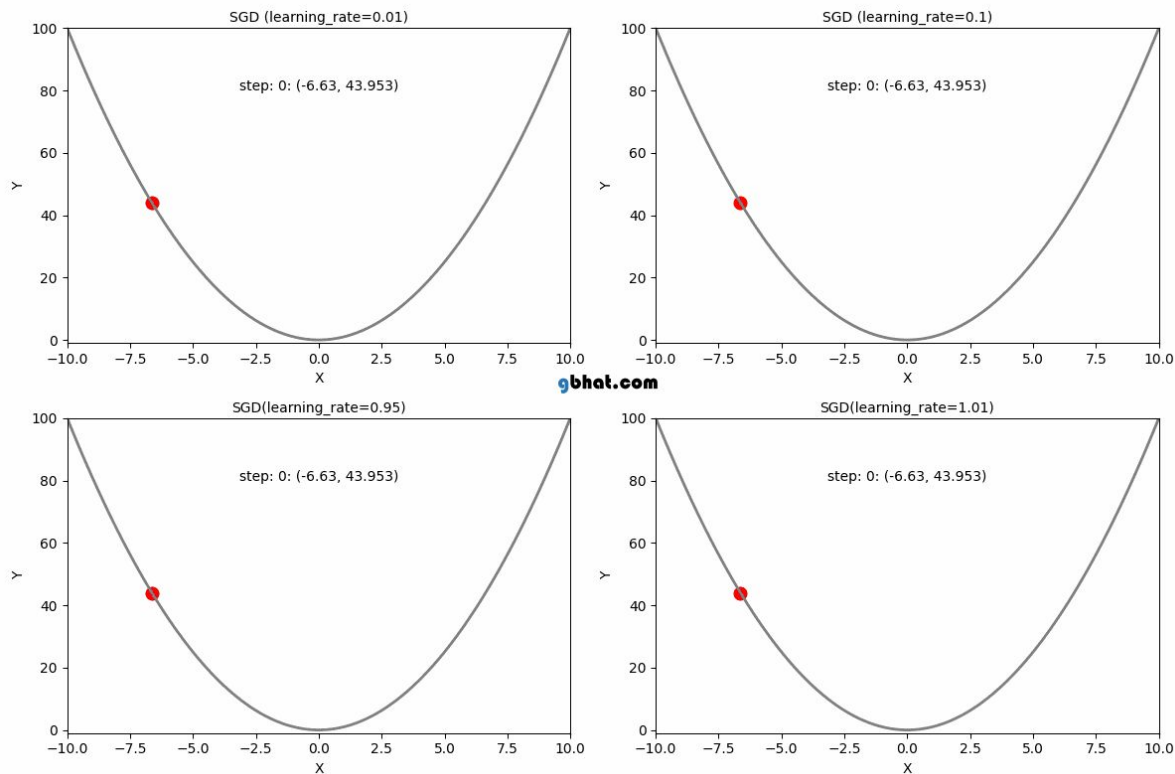
Is convergence guaranteed for all?

For confirmed convergence, the function has to be convex



Andrew Ng

Gradient Descent



Gradient Descent

- Batch Gradient Descent →

- Stable
- Slower for larger values of m

$$\theta_0 = \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$
$$\theta_1 = \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

- Stochastic Gradient Descent →

- Faster
- Noisy

$$\theta_0 = \theta_0 - \alpha (h_{\theta}(x^{(i)}) - y^{(i)})$$
$$\theta_1 = \theta_1 - \alpha (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

- Mini-batch Gradient Descent →

- Batch size is a hyperparameter

$$\theta_0 = \theta_0 - \alpha \frac{1}{B} \sum_{i \in \mathcal{B}}^b (h_{\theta}(x^{(i)}) - y^{(i)})$$
$$\theta_1 = \theta_1 - \alpha \frac{1}{B} \sum_{i \in \mathcal{B}}^b (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

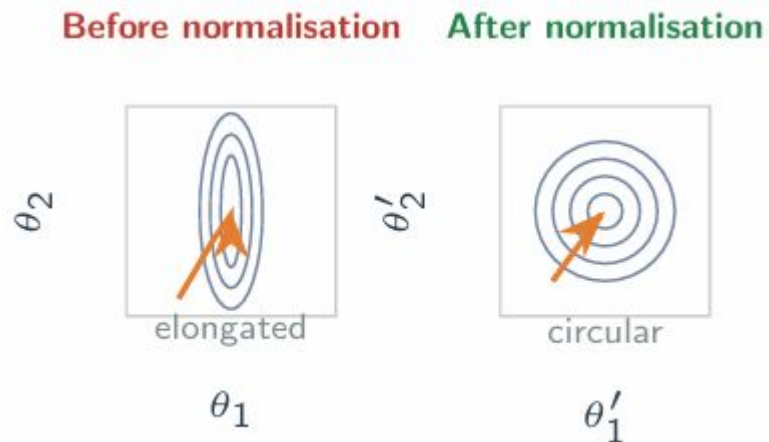
Gradient Descent

Why features should be normalized?

- Raw features can have wildly different scales
- Steps are dominated by large-scale features
- Convergence is slower

Basic normalization methods:

- Min-max normalization
- Z-Score normalization



Meme Break



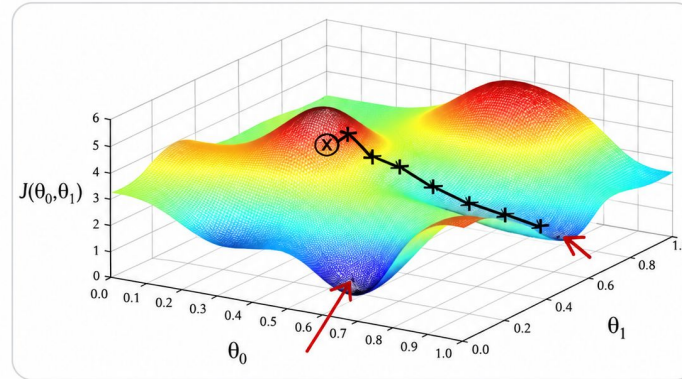
will
@willpremo



"Normalize this!!!!"

"Normalize that!!!"

Bro normalize the gradient in your gradient descent step to speed up convergence to a stationary point. lmao



Meme Break

***"Insanity is doing the same thing over
and over and expecting different results."***

-Albert Einstein

**MEANWHILE,
MACHINE LEARNING:**



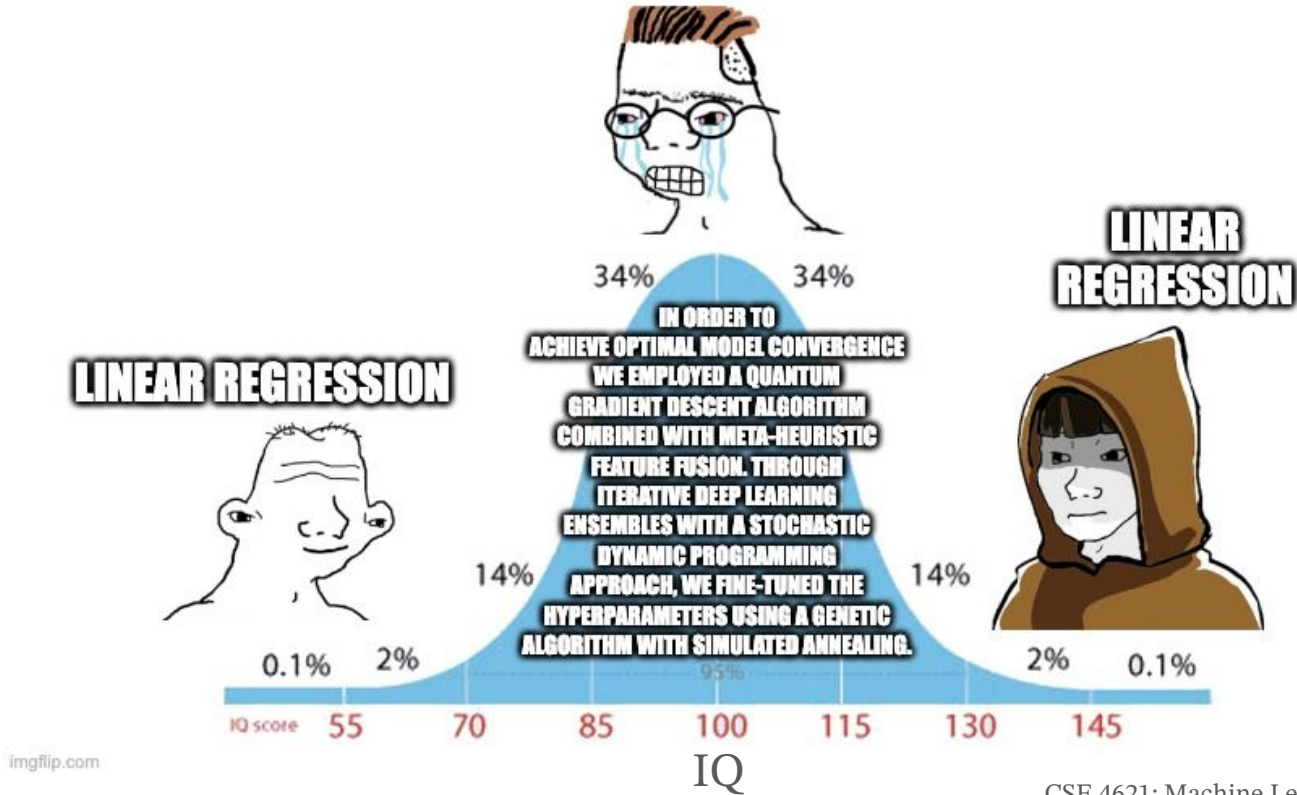
Meme Break



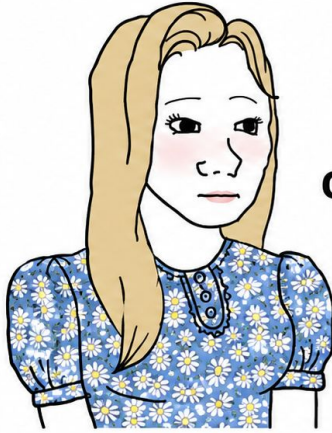
Meme Break



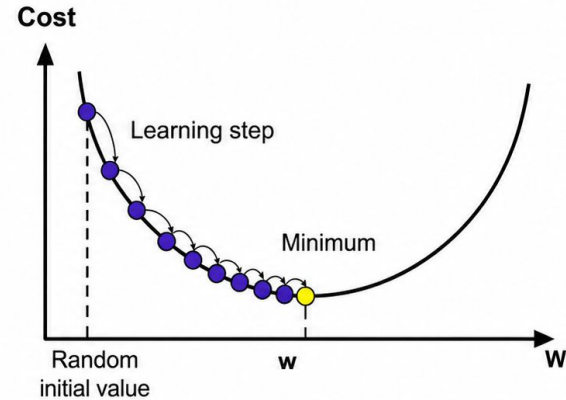
Meme Break



Meme Break



**thank you for
changing my life**



i'm literally a gradient

Textbook Chapters

- [Goodfellow] - Chapter 5.1
- [Bishop] - Chapter 3.1, 3.3

Recommended Resources

1. [Gradient Descent, Step-by-Step](#)
2. [The Hitchhiker's Guide to Hyperparameter Tuning](#)
3. [Understanding Coordinate Descent](#)

Thank you